PLACE VALUE STUDIO — PACKING BENCH & QC SPEC

Symmetric 22 White + 22 Blue-Gray Digits, 4x (+1) Carry, 4x (10) Borrow & Dual 5x2 Trays

WAREHOUSE FULFILLMENT & PACKING BENCH REFERENCE • REV a793ddd • 3x1x7U Box + 2 Inner 5x2 Trays + [123] Lid

SKU: PLACE-01-PACK

REV a793ddd

PACKING BENCH MANIFEST & REFILL REPLACEMENT PRICING

#	COMPONENT DESCRIPTION	QTY	COLOR / SPEC	CONTENT-ADDRESSED REFILL	
①	**White Digits A (4×0, 2×1–9)**	22x	PLA White	PLAC-DIG-WHT	$1.88
②	**Blue Digits B (4×0, 2×1–9)**	22x	Blue-Gray / Tan	PLAC-DIG-BLU	$1.88
③	**4x [+1] Carry & 4x [10] Borrow**	8x	Glow Yellow	PLAC-TOKN-RG	$2.25
④	**Inner 5x2 Whole Tray (L5)**	1x	Blue-Gray / Gray	PLAC-TRAY-WHL	$6.75
⑤	**Inner 5x2 Decimal Tray (L6)**	1x	Blue-Gray / Gray	PLAC-TRAY-DEC	$6.75
⑥	**3x1x7U Box (12 Sockets) & Lid**	2x	PLA White / Gray	PLAC-BOX-7U	$19.50

TOTAL: 56 VERIFIED WORKING COMPONENTS

WAREHOUSE NO LOST PIECE GUARANTEE

Individual replacement blocks are permanently stocked in Mountain View, California.
Anti-arbitrage pricing: Order single replacement components anytime at stackcalc.io/parts.
Never retire an educational classroom kit due to a single misplaced component.

PACKING QC & WATERLINE PROTOCOL

FINAL BENCH INSPECTION

- ✓ 22x White Digits: 4x [0] + 2x [1]–[9]
- ✓ 22x Blue Digits: 4x [0] + 2x [1]–[9]
- ✓ 4x [+1] Carry + 4x [10] Borrow = 8 tokens
- ✓ 2x Inner 5x2 Trays (Whole + Decimal)

PACKING WATERLINE NOTICE

56-PC PACK-UP: Seat 4 blocks in Cols 0–4 (0–9 = 40 pcs), 6 blocks in Col 5 (4x [0], 4x [+1], 4x [10] = 12 pcs), nest both 5x2 Trays over Cols 0–4, and snap Lid shut.

CALIFORNIA LAB QC AUDIT

INSPECTION: PASSED

Inspector: Station 04 • Mountain View, CA
Lot Release: 2026-Q4-ARCHIVAL
Standards: ASTM F963 / EN71 / RoHS Lead-Free

FACTORY SPECIFICATION
Harmonized Modular Mathematics System • Manufactured with Pride in California, USA.

INSTITUTIONAL DIRECT
sales@stackcalc.io

PLACE VALUE STUDIO — PACKING BENCH & QC SPEC

WAREHOUSE PACKING BENCH REFERENCE • 3D VIDEO QR • REFILL LOGISTICS

Classroom Pack-Up & Inventory Reference

SKU: PLACE-01-PACK

PACKING PROTOCOL

PACK-UP DEMO VIDEO

OPEN CAMERA & SCAN
stackcalc.io/products/place-value/
Watch 3D video demonstration

WAREHOUSE PACKING BENCH & BOX FLOOR WATERLINE AUDIT

① 1. STACK 40 DIGITS IN COLS 0–4 (LAYERS 1–4)

Place 4 blocks (2 White + 2 Blue-Gray) into each of the 10 digit sockets (Row 1: 0–4; Row 0: 5–9).

② 2. NEST BOTH 5x2 TRAYS OVER COLS 0–4 (LAYERS 5–6)

Seat the 5x2 Whole Numbers Tray (Layer 5) and 5x2 Decimal Tray (Layer 6) over Columns 0–4.

③ 3. STACK 12 BLOCKS IN COL 5 (LAYERS 1–6) & SNAP LID

Stack the 4 extra '0' blocks + 4x [+1] Carry + 4x [10] Borrow blocks 6-high in Column 5 and snap [123] Lid shut.

FULFILLMENT PORTAL
Interactive Packaging Bench Station • Warehouse Fulfillment Rev a793ddd.

REPLACEMENT MANIFEST
stackcalc.io/parts

STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

1. TACTILE CONCRETE

7.00mm 1U PITCH
Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

MOTOR KINESTHETIC
Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

CONSERVATION OF QUANTITY
Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

SELF-CORRECTING
Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

CCSS.MATH.PRACTICE.MP5
Use appropriate tools strategically.

2. HARDWARE RPN STACK

POSTFIX ARITHMETIC
[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

X, Y, Z, T REGISTERS
Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

DIGITAL TWIN OF TOWERS
Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

STEP-BY-STEP REASONING
Students observe each operator execute in real time, making compound calculations completely transparent.

CCSS.MATH.PRACTICE.MP1
Make sense of problems & persevere.

3. LEARNING CONTINUUM

GRADES K–2: FOUNDATION
Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

GRADES 3–5: FRACTIONS
Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

GRADES 6–12: ADVANCED
Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

DIFFERENTIATED TIERS
Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

CCSS.MATH.PRACTICE.MP4
Model with mathematics.

CLASSROOM CARE & SANITIZATION

1. FOOD-SAFE RECYCLED POLYMERS:
Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

2. CLASSROOM CLEANING & DISINFECTION:
Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

3. MODULAR STORAGE DETENTS:
Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

4. SOCRATIC CLASSROOM DISCUSSION:
Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

PLACE-01-PACK: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

CALIFORNIA LAB QC AUDIT CHECKLIST

[✓] MITUTOYO DIGITAL CALIPER DATUM:
All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:
Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

[✓] ZERO-RATTLE DETENT LOCK:
Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:
2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

[✓] FOOD-SAFE RECYCLED POLYMER:
100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

[✓] IN-BOX COLLATERAL PLACEMENT:
Place this reference card inside the box lid or inside the storage tray.

RPN HARDWARE COMPATIBILITY

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

TOTAL KIT INVENTORY: 56 PIECES

Verified by Assembly Tech & Machine Vision at Packing Bench.

Inspection Pass • Mountain View, CA

LIFETIME REPLACEMENT GUARANTEE

NO LOST PIECE LEFT BEHIND™

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

PORTAL: stackcalc.io/parts

DISTRICT & CLASSROOM PROCUREMENT

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io
stackcalc.io/schools

3D INTERACTIVE VIDEO DEMOS

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

VIDEOS: stackcalc.io/products

Designed & Engineered in Mountain View, California